Dhaka International University

Department of Computer Science & Engineering

Batch: Day-93

Semester: 3rd

Object-Oriented Programming Languages Lab [0613-202]

The Course Teacher, **Date: 13rd May, 2025**

Object-Oriented Programming Languages Lab [0613-202]

Subject: Seeking permission to develop CaffeineJunction [A management system for café operations] using various concepts of Object-Oriented Programming & Database Management.

Sir,

With sincere respect, I, on behalf of my team '**Pristine Falcons'** would like to state our interest to develop a management system for caffe operations containing following properties:

- **Project Title:** CAFFEINE JUNCTION

- **Features:**
- User Authentication (Login System)
- User Registration (Signup with validation)
- Product Management (Add, remove, update food items)
- Ordering System (Cart system, bill generation)
- Admin Panel (View users, modify products, view orders)
- GUI-based System (Java Swing)
- Database Connectivity (SQL-based storage)
- Security Features (Password encryption, input validation)
- Event-Driven Programming (Action Listeners, Key Listeners)

- **Objectives:**

1. Designing a simple, interactive & dynamic Management system.
2. Applying various concepts of JavaSwing & database management.
3. Integrating fluent sales process.
4. Ensuring staffs & customers satisfaction.

- **Tech Stack:** Programming Language: Java
- GUI Framework: Java Swing
- Database: SQL
- OOP Concepts: Encapsulation, Inheritance, Polymorphism, Abstraction
- **Expected Outcome:** The project will result in a functional management system capable of conducting caffe operations in simple, transparent & user-friendly process.

Our team '**Pristine Falcons'** consists of the following members who are committed to deliver the best effort in their assigned roles:

Roll No.	Name	Role
02	MOSADDEK HOSSAIN MEHEDI	Frontend & Feature Developer
17	SAIFUL ISLAM	Quality Assurance
21	HASIBUL HOQUE PROBAL	Backend & Database Developer
22	SULTANA YEASMIN SUMA	Documentation
40	AZIZUR RAHMAN ARZU	Feature Developer

We sincerely request your approval to accelerate the required follow-up actions for the successful completion of the project.

Yours sincerely,

on behalf of Pristine Falcons —

MOSADDEK HOSSAIN MEHEDI

Roll: 02

Batch: D-93

Department of Computer Science & Engineering

Submitted to:

DIDARUL ISLAM DIDAR

Lecturer

Department of Computer Science & Engineering